FABRIC CAPACITY INTELLIGENCE · MODULE 5

Understanding the data

What arrives in raw, what we do to reach silver and gold, and what the agent alerts on

We receive 60 capacity tickets with 9 fields each. **Raw** stores them exactly as exported. **Silver** types and validates them. **Gold** joins customer revenue onto them. Each ticket is then sorted into one of four outcomes, and the agent alerts on the 30 that are genuine service failures.

1 What arrives in the raw layer

60 tickets × 9 columns, stored as text exactly as ICM exported them. Nothing converted, nothing dropped, nothing corrected — so there is always an unaltered copy to compare against when a figure is questioned.

COLUMN	EXAMPLE	WHAT IT MEANS
`IncidentId`	677681988	The support ticket
`SubscriptionId`	48840256-5fa3-…	Which customer
`TenantId`	f79c09b4-e935-…	Their tenant
`Region`	westeurope	Which Azure region
`CurrentLimitCapacity`	4	What they already had
`AdditionalLimitCapacity`	83	What they asked for on top
`NewLimitCapacity`	87	What they ended up with
`DeniedDate`	2025-10-06	When refused — blank means never refused
`ApprovedDate`	2025-11-01	When granted — blank means never granted

The three capacity columns — the only tricky part of the file

Their relationship is what lets us size a shortfall, so we verified it against the data rather than assuming it:

WHEN A REQUEST WAS…	THE DATA SHOWS	HELD IN
Granted	`New = Current + Additional`	48 of 48 tickets
Never fulfilled	`New = Current` — they ended with exactly what they started with	every case

WHAT THIS ESTABLISHES

Requested = Current + Additional. **Granted** = New. Every dollar figure downstream rests on that reading, so it was confirmed against all 60 tickets before anything was built on top of it.

Two reference sheets arrive alongside the tickets

SHEET	ROWS	WHAT IT GIVES US
`Subscription_Reference`	26	Customer → tier and annual revenue. **Placeholder data today.**
`Expected_Classifications`	60	The hand-labelled correct answer for every ticket — used to test the classifier before trusting it

2 Raw → Silver: making it trustworthy

Same 60 tickets, now typed and validated. Six operations:

#	WHAT WE DO	WHY IT IS NECESSARY
1	Trim text; blank out literal `"nan"` and `"None"`	Spreadsheet exports leave those as actual text, which would be read as real values
2	Parse dates into proper UTC timestamps	You cannot subtract two pieces of text. Anything unreadable becomes empty — never a guess
3	Convert capacity fields to numbers	Arithmetic is impossible on text
4	Remove duplicate `IncidentId`	Keeps the first; the rest are listed by ID, not silently discarded
5	Drop rows with no region, or with *neither* date	Nothing meaningful can be said about them
6	Flag approvals dated *before* their denial	**Flagged, not dropped** — given their own category so a reviewer sees them

> RESULT ON THIS EXTRACT
>
> **60 of 60 tickets usable — no problems found.** No duplicates, no missing regions, no rows without dates, no inverted date pairs, no negative capacities.

Every check produces both a count and the specific incident IDs. That report prints on every run and is carried into the Teams message, so rows can never disappear quietly and a shrinking total is always explained.

3 Silver → Gold: adding the business context

One operation: **join each ticket to its customer's revenue**, matching on `SubscriptionId`.

LAYER	SHAPE	COLUMNS ADDED
Silver	60 × 9	—
Gold	60 × 12	`SubscriptionTier`, `ARR_USD`, join indicator

Gold exists so the analysis has both halves of the story in a single row: the *technical* shortfall from the ticket, and the *business value* from the revenue reference. Units of capacity cannot be converted into dollars until those sit side by side.

> WHEN A CUSTOMER HAS NO REVENUE RECORD
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> They contribute $0 rather than failing the run, and are named in the data-quality report — so a gap in the reference data looks like a gap, instead of being mistaken for good news.

4 What the agent actually alerts on

Not all 60 tickets — only the 30 that represent genuine failures. Each Gold row is sorted by comparing its two dates:

OUTCOME	TICKETS	ALERTED ON?	REASONING
Denied, then approved **more than 48h later**	18	**Yes**	Customer carried the shortfall while waiting
Denied, **never approved**	12	**Yes**	Still without the capacity; exposure still growing
Denied, approved within 48h	15	No	Normal turnaround, not a service failure
Never denied at all	15	No	Nothing went wrong
Total	**60**	**30 alerted**	

From 60 tickets to one message

The 30 failures are priced into dollars, grouped by region, ranked by exposure, and the top three regions each get one specific recommendation. Nothing is executed automatically — every message is held for a person to approve.

5 Two things worth saying before being asked

Both concern how far the numbers can currently be trusted. Stating them unprompted is what makes the rest credible.

> 1 · THE REVENUE FIGURES ARE PLACEHOLDER
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> The customer ARR reference is synthetic. The **ranking of regions holds**, because it depends on relative customer size — but the absolute dollar amounts will change when the real revenue reference arrives. Every message carries this caveat on its face until it does.

> 2 · "NEVER FULFILLED" IS PROBABLY OVERSTATED
>
> ICM gives us no ticket status, so we cannot separate *rejected permanently* from *still being worked* — both are currently counted as failures. That affects **12 of the 30** flagged requests, and because open tickets accrue exposure daily, it inflates the headline. A status column from ICM is the single change that would most improve the honesty of the number.

Figures from the synthetic ICM extract, 60 tickets, as of 2026-01-28. Revenue figures illustrative. Full detail in Module5_Reference.pdf.